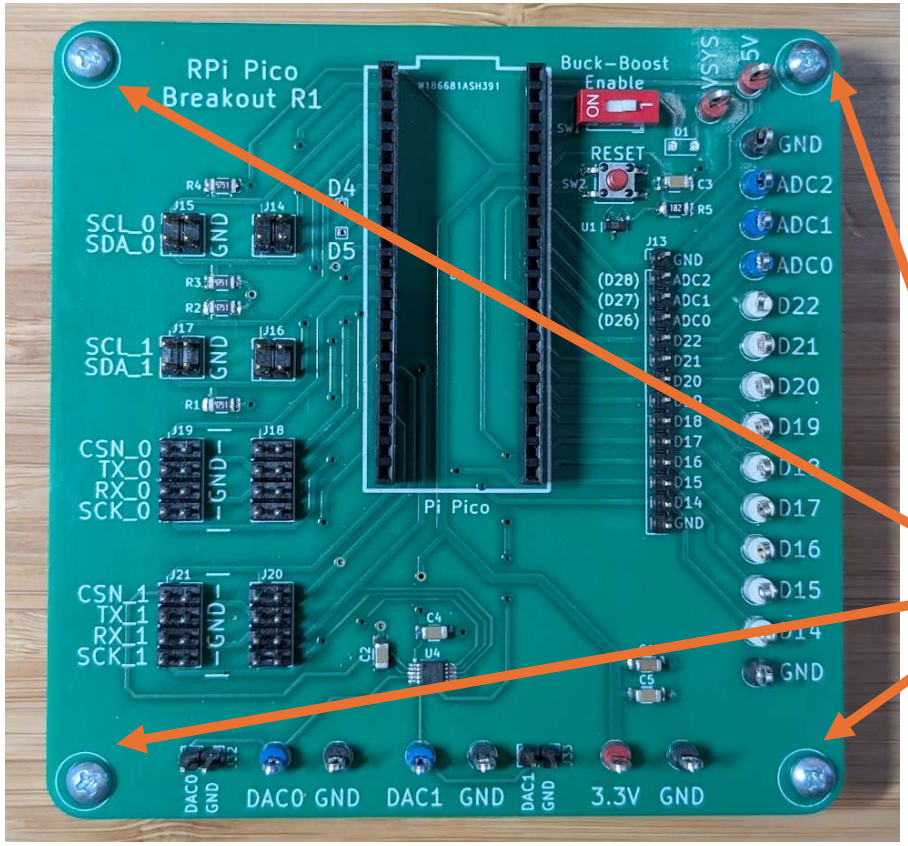
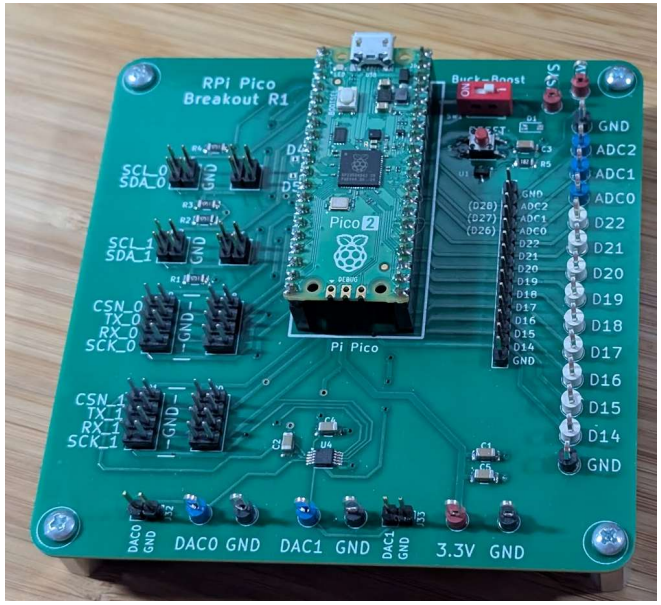
Class #1

Step 1



1. Assemble stand-offs (x4)

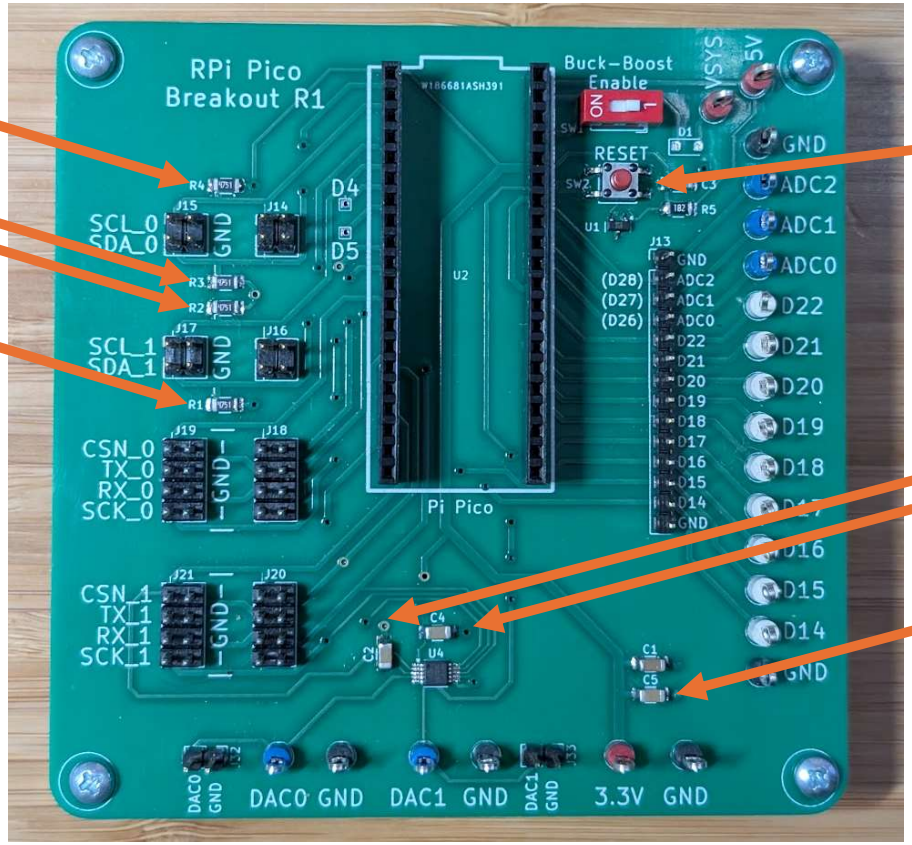
Finished Project



KiCAD PCB files, schematics, docs can be found at:
https://github.com/KnoxMakers/Electronics/tree/master/Classes/2025-12-06%20RPI%20Pico%20Class

Steps 2-4

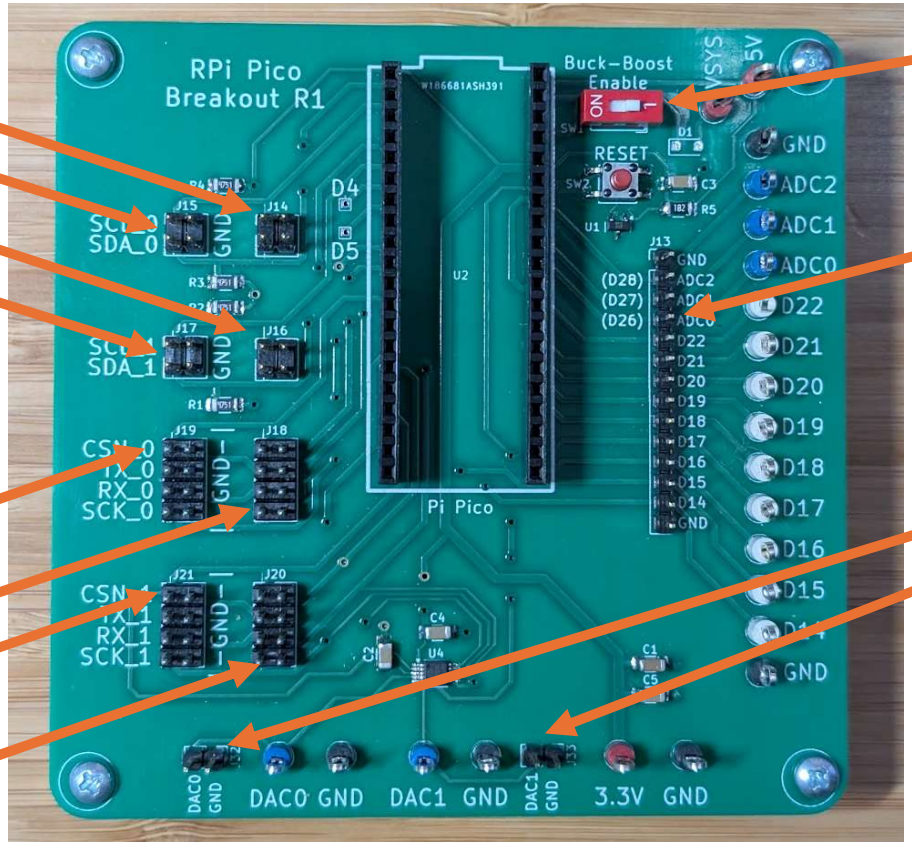
2. Solder SMT Resistors (x4)



3. Solder SMT Switch (x1)

4. Solder SMT Capacitors (x3)

Steps 5-9



6. Solder 2x2 Headers (x4)

5. Solder 4x2 Headers (x4)

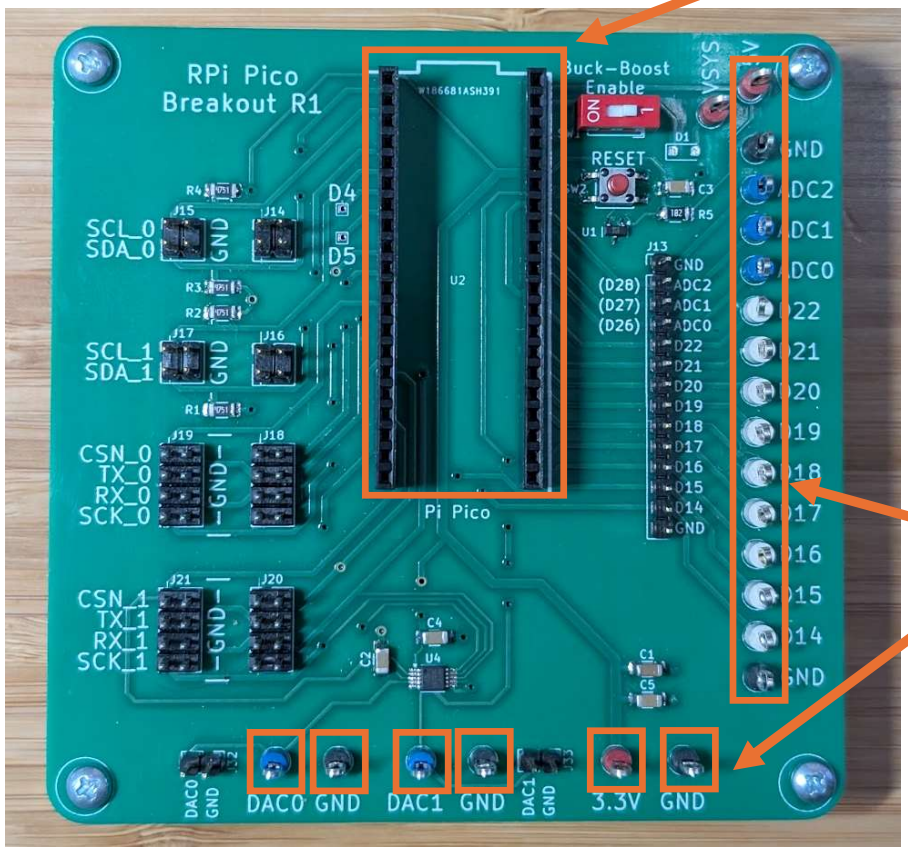
8. Solder Switch (x1)

7. Solder 14x1 Header (x1)

9. Solder 2x1 Headers (x2)

Steps 6-7

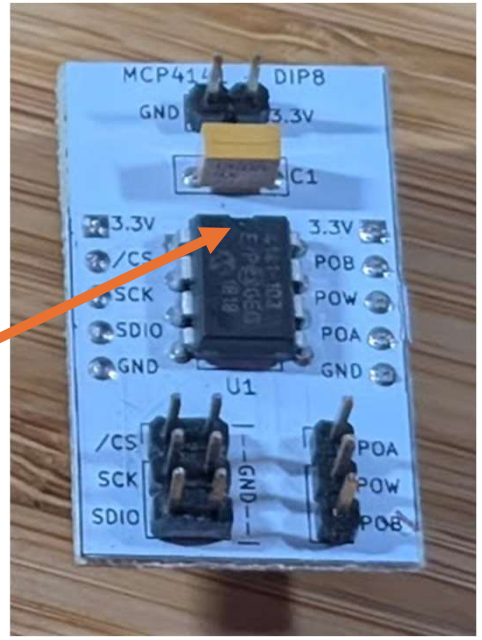
7. 20-Pin Headers (x2)



6. Solder Keystone Test Points (x21)

Digipot Board

Note location of notch



Class #2

Class #2 Overview

Files from Class

https://github.com/KnoxMakers/Electronics/tree/master/Classes/2025-12-06%20RPI%20Pico%20Class

Priority #1

- Learn to flash the Pico
- Learn to correctly hook up the breakouts to the Pico
- Verify your hardware is working

Priority #2 (as time allows)

- Explain high level Arduino sketch organization
- Explain I2C and SPI bus communications
- Walk through select areas of the code

We won't walk through the code in detail, explain C++, etc. That is left as an exercise on your own, if you desire.

If there is interest, we could hold some intro programming classes.

As always, the forums are available for help and Tuesday nights are great.

Note: Rob may do a micropython class or Lightening Talk related to the Pico and hardware used in this class, be on the lookout.

Setup Arduino IDE

(This has already been done)

1. File->Preferences

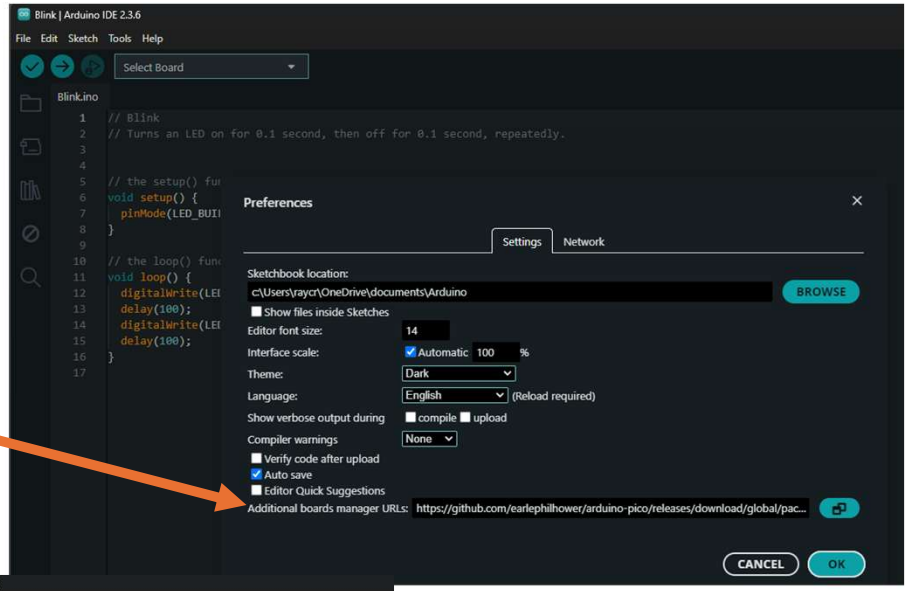
Add the Pico Board to the additional board manager URLs

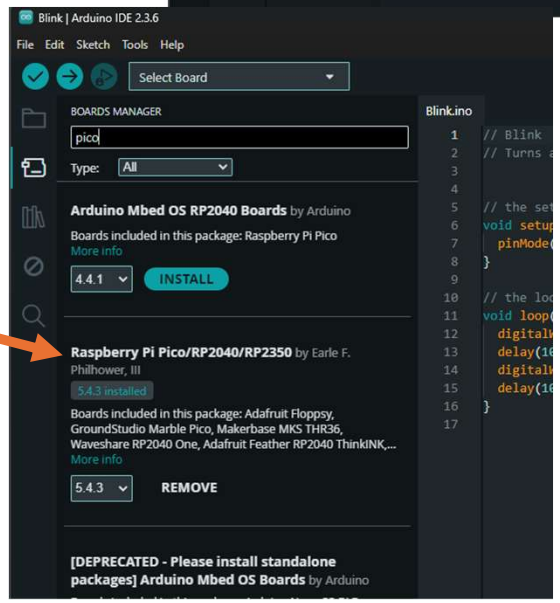
https://github.com/earlephilhower/Arduino-pico/releases/download/global/package_rp2040_index.json

2. Tools->Boards->Board Manager

Search for “Earle”

Install the “Raspberry Pi Pico/RP2040/RP2350 by Earle F. Philhower, III





Flash the Blink sketch

1. File->Open
Navigate to Downloads\PicoClass\Blink and click on “Blink.ino”

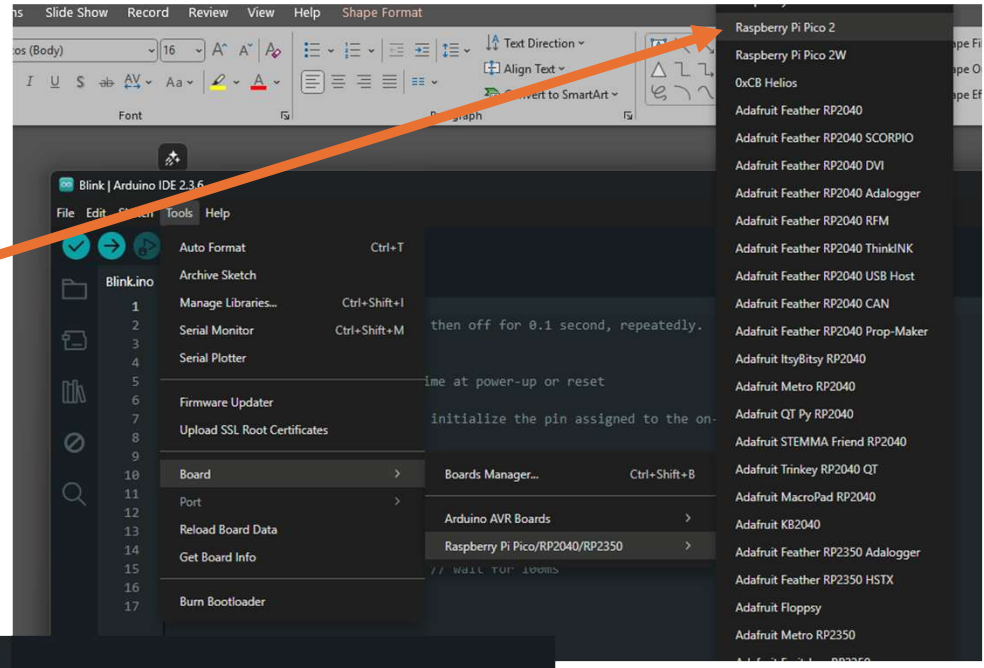
2. Tools->Board->Raspberry Pi Pico/RP2040/RP2350->Raspberry Pi Pico 2

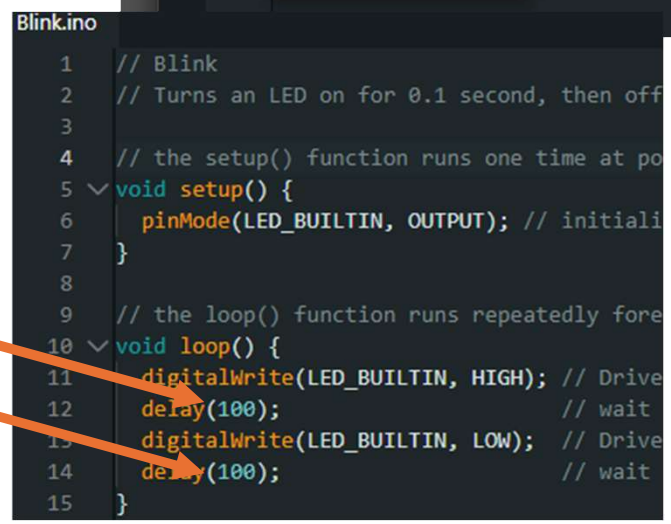
3. Tools->Port->UF2 Board or some other port

4. Sketch->Upload
- Will compile and, if no errors, flash the Pico

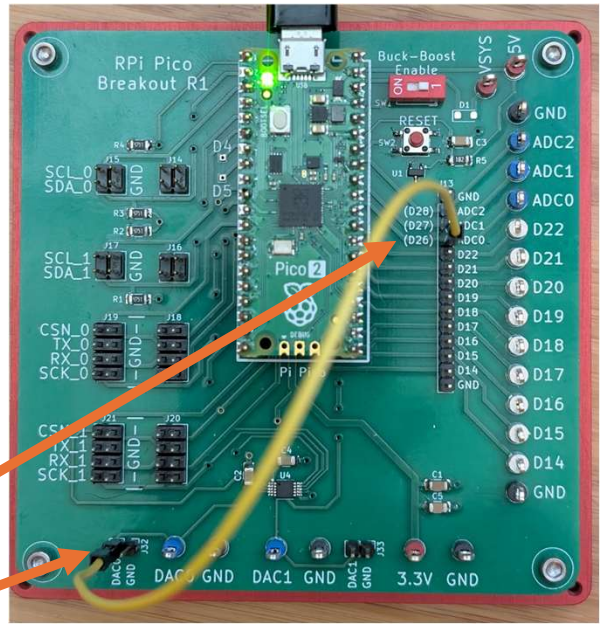
5. Look for the LED on the Pico to flash

6. Change these numbers and Upload





DAC_Test Sketch



1. File->Open
Navigate to Downloads\ PicoClass\DAC_Test and click on "DAC_Test.ino"

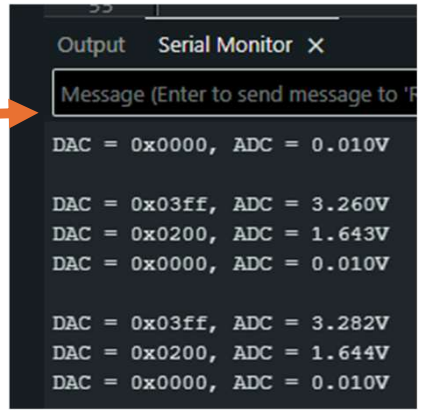
2. Tools->Board->Raspberry Pi Pico/RP2040/RP2350->Raspberry Pi Pico 2

3. Tools->Port->something

4. Sketch->Upload
   Will compile and, if no errors, flash the Pico

5. Connect jumper wire from DAC0 to ADC0

6. Tools->Serial Monitor
   Observe the DAC being set and the ADC reading the voltage
   0x000 is minimum voltage (0V)
   0x3FF is maximum voltage (3.3V)



Temp Sensor Sketch

1. File->Open
Navigate to Downloads\ PicoClass\ MP1075_Breakout_Test and click on “TMP1075_Breakout_Test.ino”

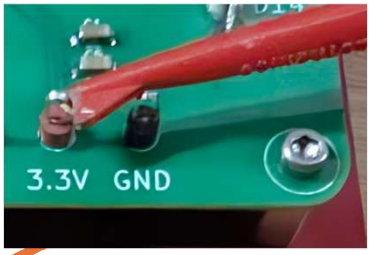
2. Tools->Board->Raspberry Pi Pico/RP2040/RP2350->Raspberry Pi Pico 2

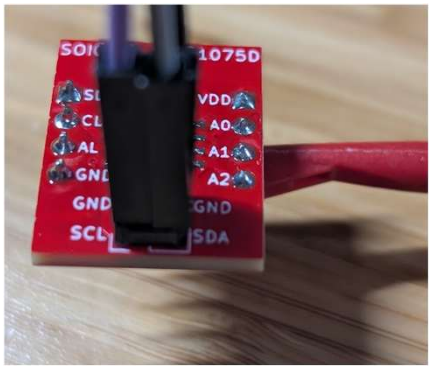
3. Tools->Port->something

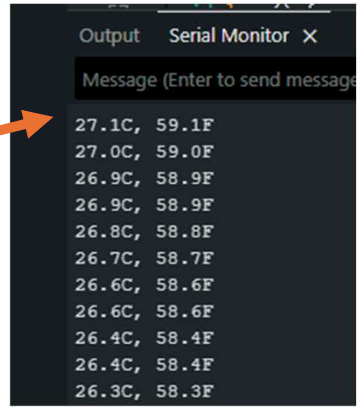
4. Sketch->Upload
Will compile and, if no errors, flash the Pico

5. Connect jumper wires:
GND to GND (any GND to any GND)
SDA_1 to SDA
SCL_1 to SCL
3.3V to VDD (back of PCB)

6. Tools->Serial Monitor







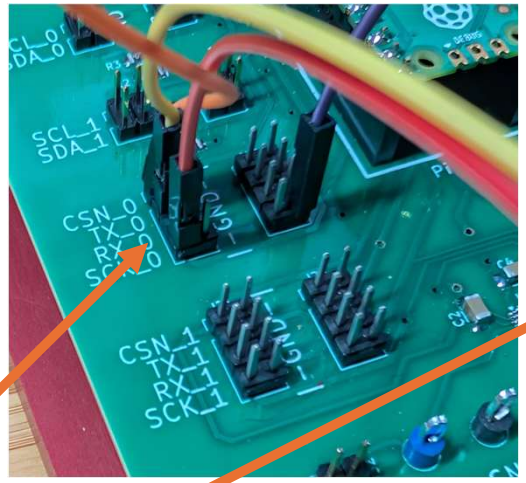
Digipot Sketch

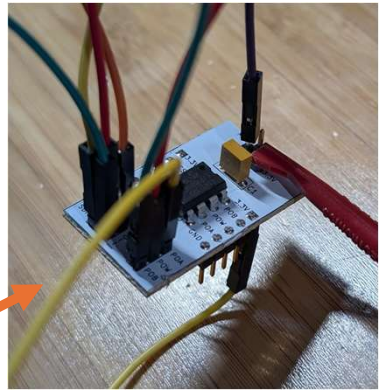
1. File->Open
Navigate to Downloads\ PicoClass\ MCP4141_Breakout_Test and click on the .ino file

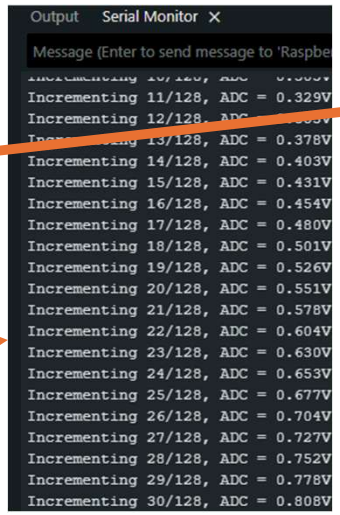
2. Sketch->Upload
Will compile and, if no errors, flash the Pico

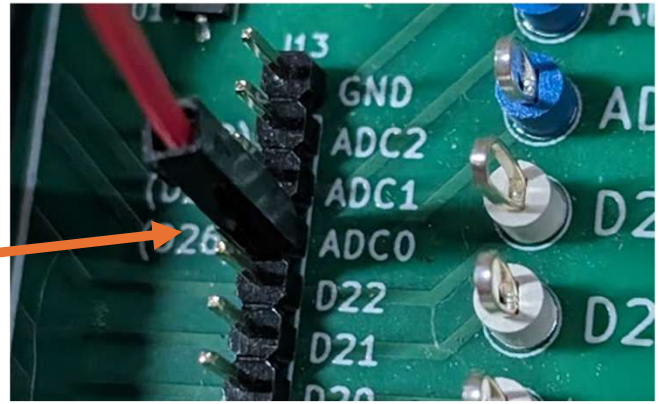
3. Connect jumper wires:
Pico to Breakout:
GND to GND (any GND to any GND)
3.3V to 3.3V
CSN_0 to /CS
TX_0 to SDIO
SCK_0 to SCK
ADC0 to POW
Breakout to Breakout:
3.3V (back of PCB) to POA (front of PCB)
GND to POB

4. Terminal

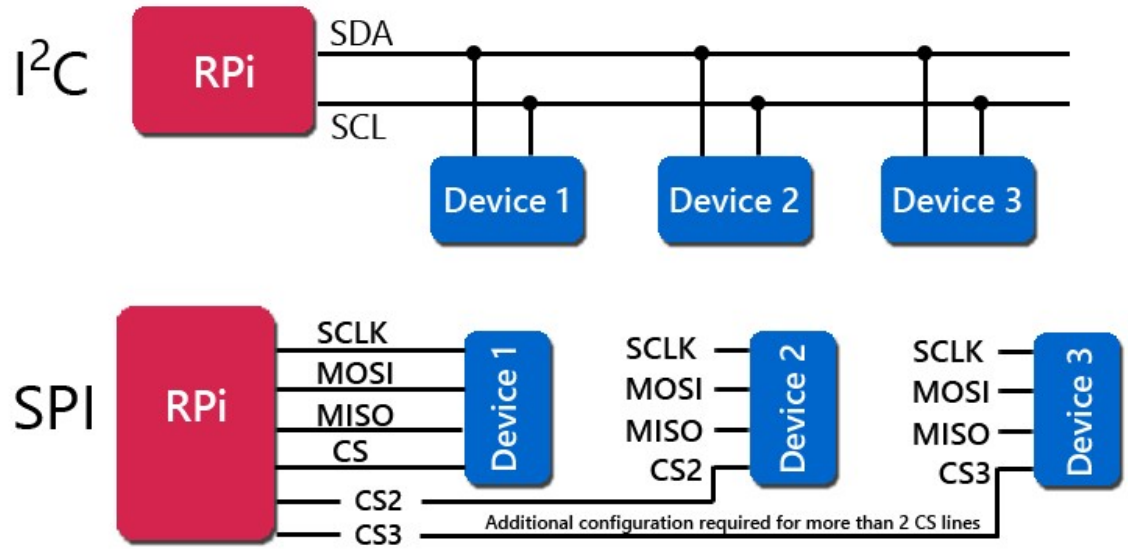








I2C vs SPI



I2C:

Data rate typically slower (100kbps, 400kbps, 1Mbps)
RPi get an acknowledgement from slave
Only 2 wires for comms to many devices

SPI:

Data rate typically faster (1Mbps to 62.5Mbps)
RPi gets no ack (can talk into space and not know it)
Requires 3 wires common to all devices + 1 chip select for each